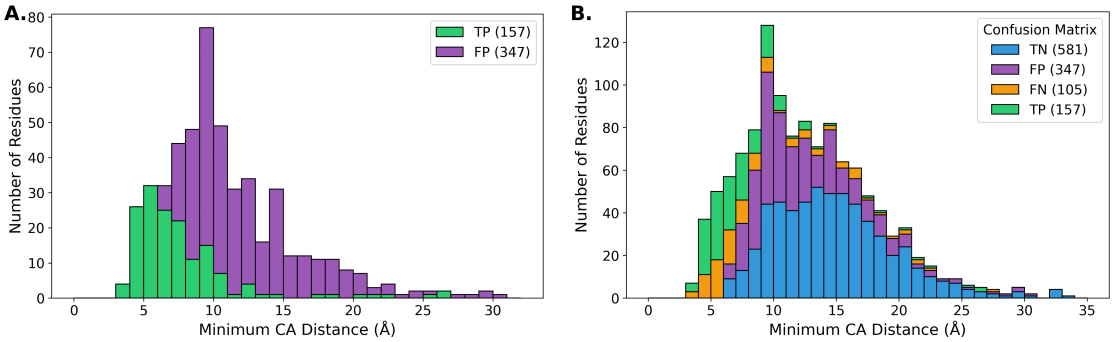
## SI Table S2. Data for Hydrolase Receptors

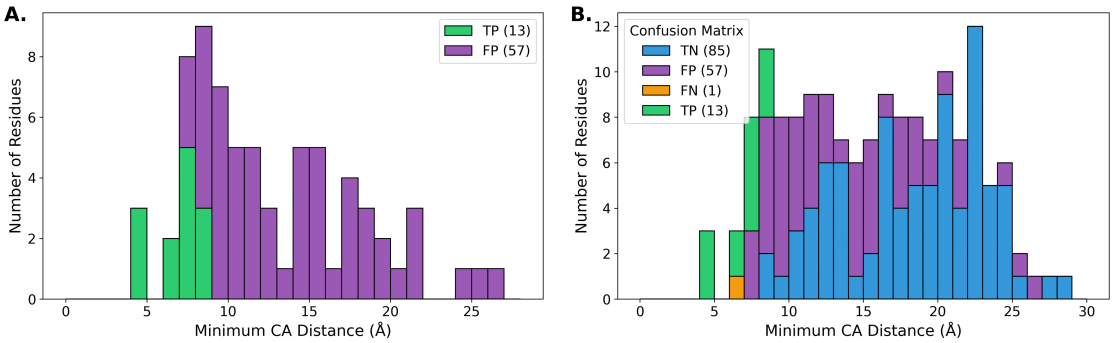
| apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC | apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2LBM | 15001 | 17569 | 0.60 | 0.46 | 7QCX | 1D5G | 34688 | 4516 | 0.27 | 0.22 |
| | 6FDP | 19757 | 34223 | 0.48 | 0.43 | | 6OQJ | 4276 | 30605 | 0.32 | 0.22 |
| 2GS0 | 2LOX | 6225 | 18229 | 0.47 | 0.42 | | 6OQJ | 30599 | 30605 | 0.38 | 0.17 |
| | 2PLD | 5318 | 5310 | 0.46 | 0.40 | | 1VJ6 | 5131 | 6060 | 0.29 | 0.15 |
| | 2RQG | 4915 | 10019 | 0.43 | 0.30 | | 1VJ6 | 5131 | 6060 | 0.29 | 0.15 |
| | 6GC3 | 17173 | 34260 | 0.29 | 0.28 | | 1ONV | 27288 | 5685 | 0.34 | 0.12 |
| | 2LOZ | 10318 | 17364 | 0.38 | 0.27 | 2KQ3 | 2KHS | 16585 | 15357 | 0.31 | 0.11 |
| | 1CF4 | 18251 | 4700 | 0.49 | 0.27 | | 2LE8 | 16396 | 17702 | 0.21 | 0.10 |
| | 2KC8 | 16066 | 16065 | 0.52 | 0.27 | | 2HUG | 6592 | 7241 | 0.35 | 0.09 |
| | 2KWI | 15524 | 15525 | 0.47 | 0.26 | | 2LTO | 10276 | 18490 | 0.14 | 0.09 |
| | 6R5G | 28070 | 34384 | 0.39 | 0.25 | 2L5U | 2L75 | 17285 | 17344 | 0.44 | 0.02 |
| | 2K7L | 27288 | 15919 | 0.39 | 0.25 | | 7NQC | 51289 | 34608 | 0.34 | 0.01 |
| | 2KYL | 15972 | 16967 | 0.41 | 0.23 | | 2RMK | 5511 | 11010 | 0.18 | -0.02 |
| | 2KWI | 15230 | 15525 | 0.36 | 0.23 | 2JNS | 6BGG | 15125 | 30367 | 0.14 | -0.08 |
| | 2JZI | 51289 | 15624 | 0.43 | 0.22 | | | | | | |




**SI Fig. S1. Confusion Matrix Histograms for Hydrolase Receptors**

## SI Table S3. Data for Isomerase Receptors

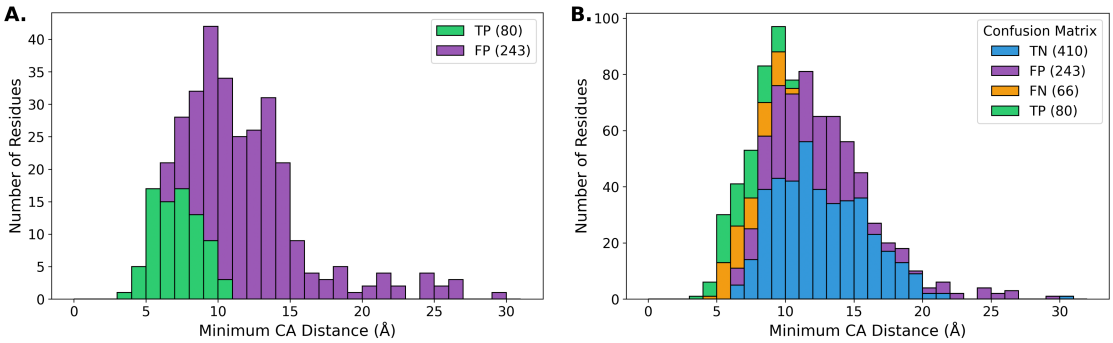
| apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC | apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2MS4 | 2208 | 25104 | 0.29 | 0.29 | | 7ZEY | 16983 | 34727 | 0.48 | 0.18 |
| | 8SG2 | 27579 | 31080 | 0.44 | 0.23 | | 2RSE | 50325 | 11471 | 0.05 | -0.02 |
| | 7ZEY | 34724 | 34727 | 0.36 | 0.22 | | 2RSE | 16931 | 11471 | 0.00 | -0.16 |




**SI Fig. S2. Confusion Matrix Histograms for Isomerase Receptors**

## SI Table S4. Data for Oxidoreductase Receptors

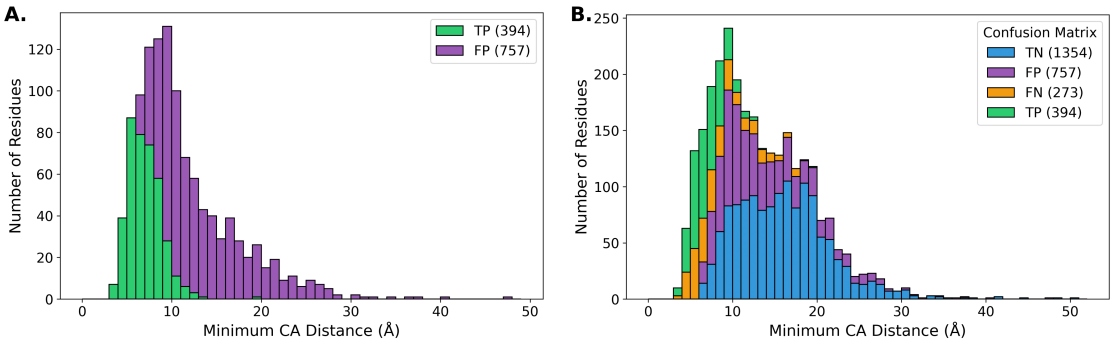
| apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC | apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2JNS | 6BNH | 15125 | 30373 | 0.48 | 0.36 | | 2LL7 | 51289 | 18028 | 0.33 | 0.17 |
| | 2KGI | 16209 | 16210 | 0.62 | 0.30 | | 2M56 | 17415 | 19038 | 0.10 | 0.16 |
| | 7KLR | 30808 | 30809 | 0.55 | 0.29 | | 2M56 | 4154 | 19038 | 0.12 | 0.04 |
| | 2MG5 | 51289 | 19586 | 0.44 | 0.27 | | 2MNZ | 19913 | 19914 | 0.44 | 0.02 |
| | 2IPA | 6075 | 15028 | 0.18 | 0.23 | | 2N8J | 51289 | 25852 | 0.19 | -0.01 |
| | 2LL6 | 51289 | 18027 | 0.39 | 0.21 | | 5TP6 | 51289 | 30196 | 0.21 | -0.03 |




**SI Fig. S3. Confusion Matrix Histograms for Oxidoreductase Receptors**

## SI Table S5. Data for Transferase Receptors

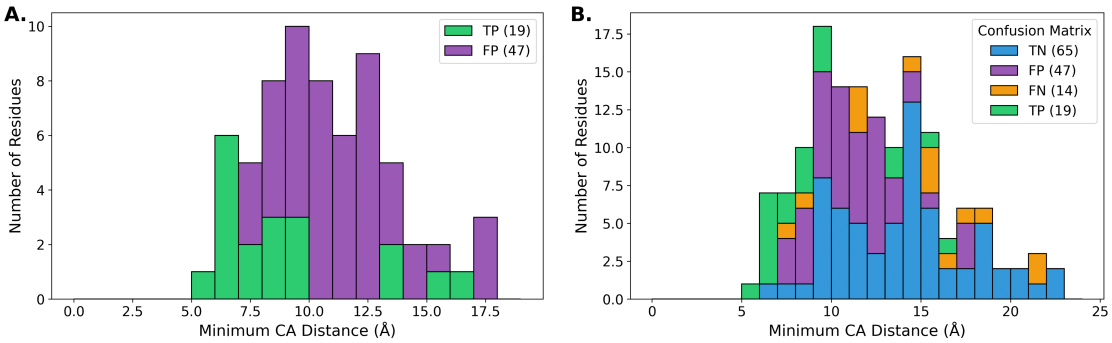
| apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC | apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2N9P | 25913 | 25914 | 0.72 | 0.58 | 2L0I | 5M9D | 17044 | 34058 | 0.33 | 0.18 |
| 7JMY | 7JYN | 30782 | 30790 | 0.53 | 0.43 | | 2YS5 | 11095 | 11094 | 0.35 | 0.18 |
| | 2N8T | 25866 | 25865 | 0.64 | 0.42 | 6VED | 2L3R | 30703 | 17200 | 0.30 | 0.18 |
| | 2C52 | 15398 | 6874 | 0.56 | 0.41 | | 7ZEY | 16983 | 34727 | 0.48 | 0.18 |
| | 2PLD | 5318 | 5310 | 0.46 | 0.40 | | 5J8H | 51289 | 30063 | 0.43 | 0.18 |
| 5AHT | 2KPZ | 25349 | 16574 | 0.55 | 0.40 | | 5IAY | 30703 | 30019 | 0.22 | 0.17 |
| 1W1F | 1WA7 | 6261 | 6456 | 0.51 | 0.38 | | 2LXS | 6095 | 18694 | 0.32 | 0.17 |
| 2LSY | 2N1G | 18455 | 25559 | 0.45 | 0.37 | | 2LVO | 18610 | 18582 | 0.32 | 0.16 |
| | 2KWU | 17769 | 16880 | 0.40 | 0.37 | | 2MV7 | 18094 | 19516 | 0.38 | 0.15 |
| 2KM4 | 2L0I | 16411 | 17044 | 0.36 | 0.36 | | 2MRE | 18610 | 25070 | 0.28 | 0.13 |
| | 2KWV | 17769 | 16885 | 0.39 | 0.34 | | 5JYV | 30092 | 30093 | 0.31 | 0.13 |
| 2LSG | 2LSJ | 18431 | 18433 | 0.35 | 0.34 | | 6QXZ | 27250 | 27251 | 0.42 | 0.13 |
| | 2MC6 | 19428 | 19429 | 0.47 | 0.33 | | 2H7D | 15792 | 7150 | 0.24 | 0.11 |
| | 5JYV | 30091 | 30093 | 0.48 | 0.33 | | 9C5E | 18610 | 31177 | 0.30 | 0.11 |
| 2LSY | 2LSK | 18455 | 18434 | 0.38 | 0.32 | | 2M86 | 15945 | 19230 | 0.25 | 0.10 |
| | 2LVO | 18581 | 18582 | 0.50 | 0.30 | | 2N1A | 6304 | 25553 | 0.10 | 0.10 |
| 5AHT | 2KQ0 | 25349 | 16575 | 0.50 | 0.30 | | 2KA6 | 4789 | 16015 | 0.51 | 0.09 |
| | 2MRE | 25071 | 25070 | 0.61 | 0.29 | | 2LTO | 10276 | 18490 | 0.14 | 0.09 |
| | 2OJ2 | 4122 | 6581 | 0.37 | 0.28 | | 2K7A | 5461 | 16809 | 0.21 | 0.09 |
| | 5VF0 | 17769 | 30276 | 0.31 | 0.27 | | 2K7A | 11018 | 16809 | 0.33 | 0.08 |
| | 1CF4 | 18251 | 4700 | 0.49 | 0.27 | | 2KJE | 4789 | 16318 | 0.45 | 0.07 |
| 2JNS | 2NCZ | 15125 | 26041 | 0.40 | 0.25 | | 2MZD | 34231 | 25484 | 0.35 | 0.06 |
| | 2LGK | 17813 | 17812 | 0.52 | 0.25 | | 2RSY | 7141 | 11508 | 0.30 | 0.05 |
| | 2MOW | 17173 | 19954 | 0.21 | 0.25 | | 2G35 | 15792 | 7061 | 0.17 | 0.04 |
| | 2MBB | 15410 | 19394 | 0.43 | 0.24 | | 2MSR | 34179 | 25130 | 0.12 | 0.00 |
| 2L0I | 5LVF | 17044 | 34041 | 0.31 | 0.24 | | 2RSE | 50325 | 11471 | 0.05 | -0.02 |
| 1U2N | 1R8U | 6268 | 5987 | 0.56 | 0.24 | | 2RMK | 5511 | 11010 | 0.18 | -0.02 |
| | 1I5H | 26698 | 4963 | 0.40 | 0.23 | | 2LGG | 17813 | 17808 | 0.37 | -0.03 |
| | 2KYL | 15972 | 16967 | 0.41 | 0.23 | | 2I94 | 5332 | 7293 | 0.17 | -0.03 |
| | 8SG2 | 27579 | 31080 | 0.44 | 0.23 | | 2N83 | 25828 | 25833 | 0.29 | -0.03 |
| 1U2N | 1L8C | 6268 | 5327 | 0.58 | 0.22 | | 2K8F | 34231 | 15944 | 0.22 | -0.05 |
| | 7ZEY | 34724 | 34727 | 0.36 | 0.22 | 2JNS | 2ND1 | 15125 | 26043 | 0.23 | -0.06 |
| 1U2N | 2KA4 | 6268 | 16014 | 0.55 | 0.21 | | 2LV6 | 51289 | 18556 | 0.35 | -0.07 |
| | 2MH0 | 34231 | 19610 | 0.33 | 0.21 | | 2N83 | 25883 | 25833 | 0.29 | -0.10 |
| | 2MPS | 6612 | 18876 | 0.34 | 0.20 | | 9C5E | 18642 | 31177 | 0.35 | -0.11 |
| | 6U19 | 16620 | 27875 | 0.43 | 0.20 | | 2RSE | 16931 | 11471 | 0.00 | -0.16 |
| | 2LAS | 50128 | 25553 | 0.28 | 0.19 | | 2JMF | 6262 | 15016 | 0.13 | -0.17 |




**SI Fig. S4. Confusion Matrix Histograms for Transferase Receptors**

## SI Table S6. Data for Translocase Receptors

| apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC | apo_pdb | holo_pdb | apo_bmrb | holo_bmrb | F1 | MCC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2KWI | 15524 | 15525 | 0.47 | 0.26 | | 2KNE | 51289 | 16465 | 0.36 | 0.13 |
| | 2KWI | 15230 | 15525 | 0.36 | 0.23 | | | | | | |




**SI Fig. S5. Confusion Matrix Histograms for Translocase Receptors**